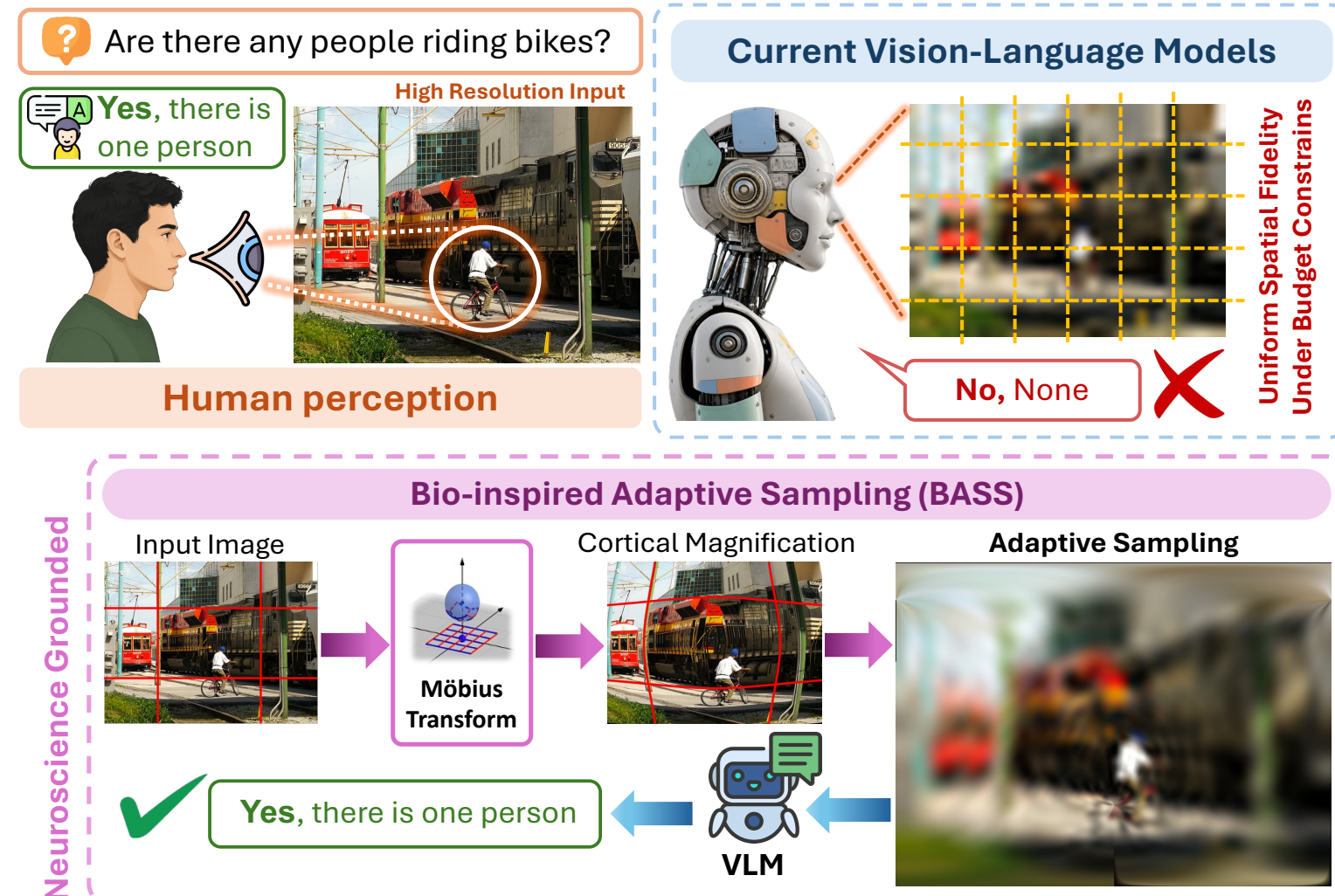




Motivation



Current VLMs assume uniform spatial fidelity, allocating equal computation to all pixels. This leads to **wasted resources on uninformative regions** and loss of critical details.

In contrast, human vision is inherently **Adaptive, Selective, Efficient** and **Task-Driven** using Cortical Magnification & Saccadic Eye Movement

Research Question.

Can biologically inspired sampling strategies VLMs to achieve better reasoning than conventional uniform sampling under limited pixel budgets?

Contributions

01

A bio-inspired training-free adaptive sampling framework for efficient VLM perception.

02

A Möbius transformation-based sampling strategy for non-uniform pixel allocation.

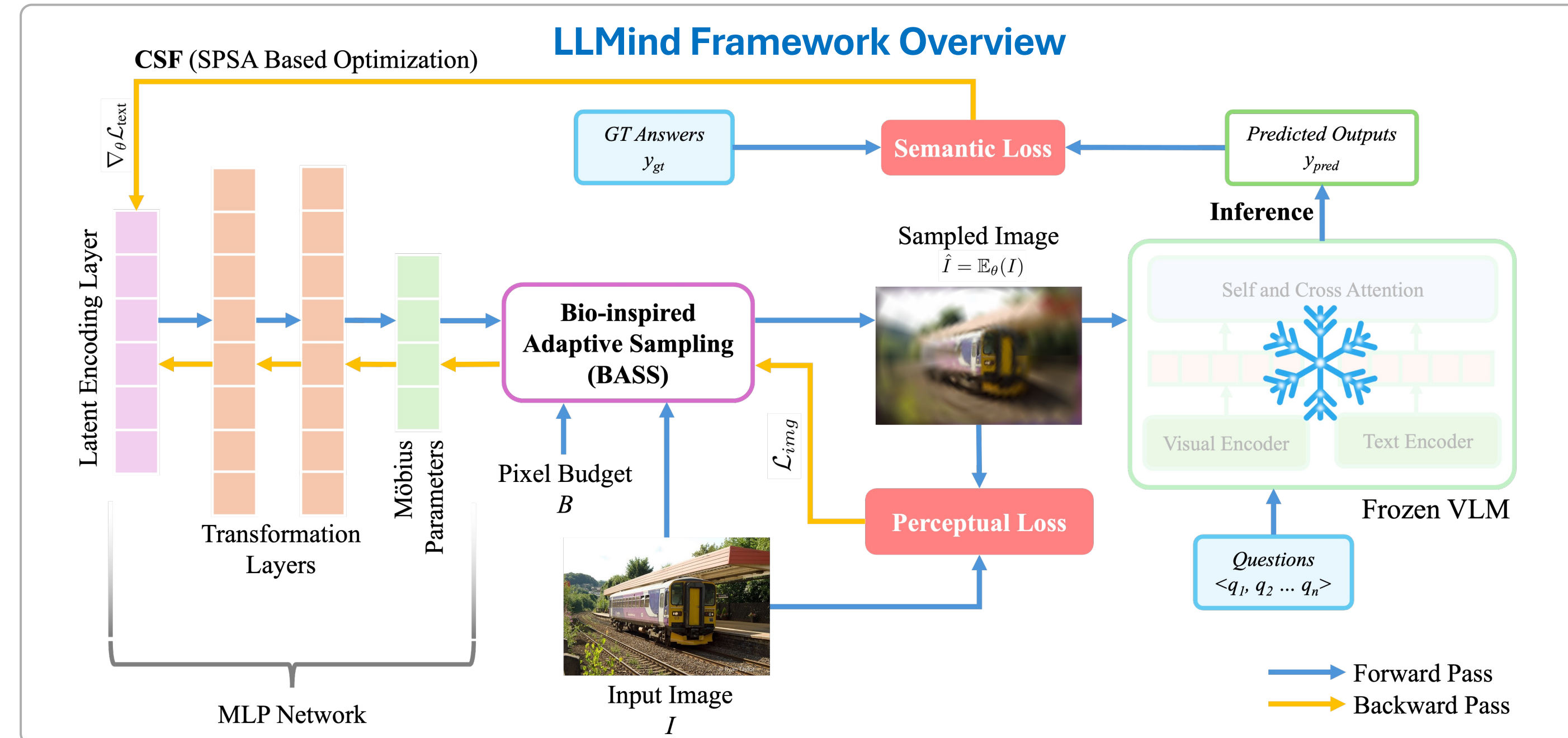
03

A model agnostic, closed-loop semantic feedback (CSF) for test-time alignment.

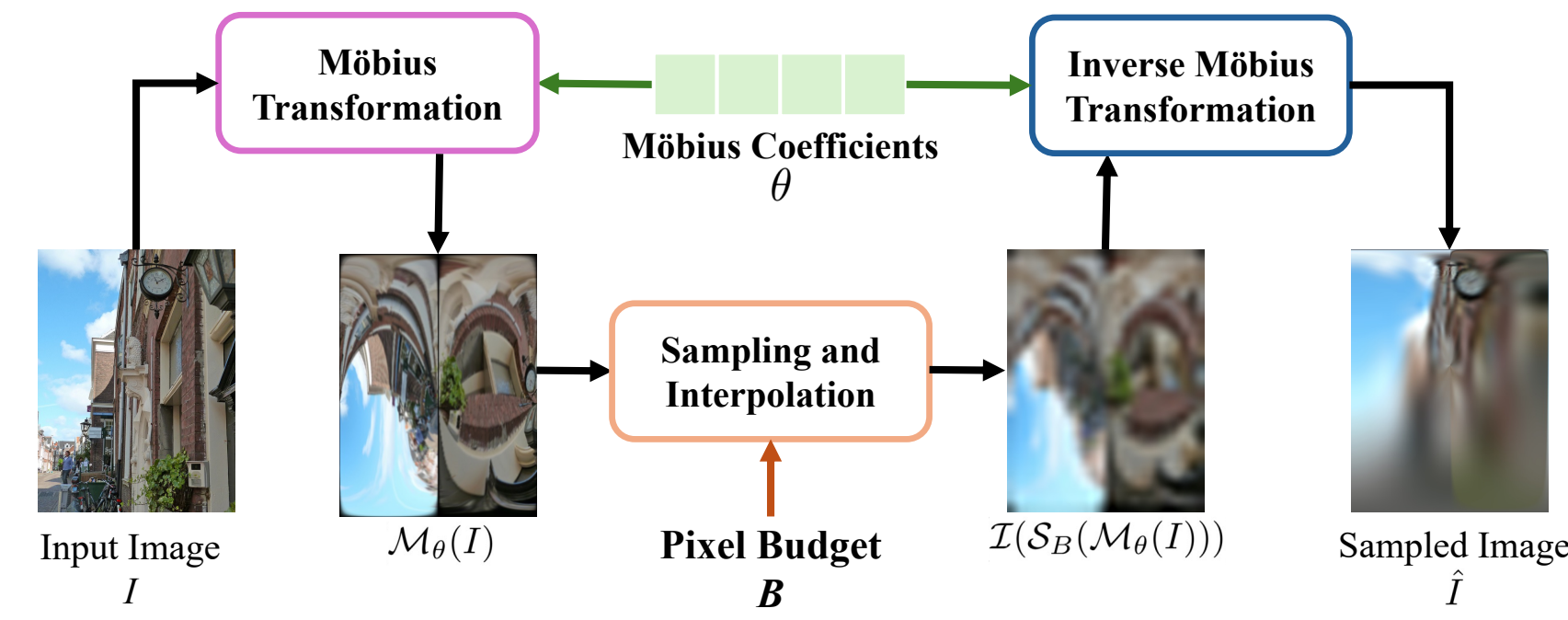
04

Consistent VQA gains under extreme pixel budget constrains across diverse settings.

Methodology



Bio-inspired Adaptive Sampling Strategy (BASS)



BASS models **cortical magnification** by warping images toward task-relevant regions, while CSF mimics **saccadic eye movements** through iterative refinements.

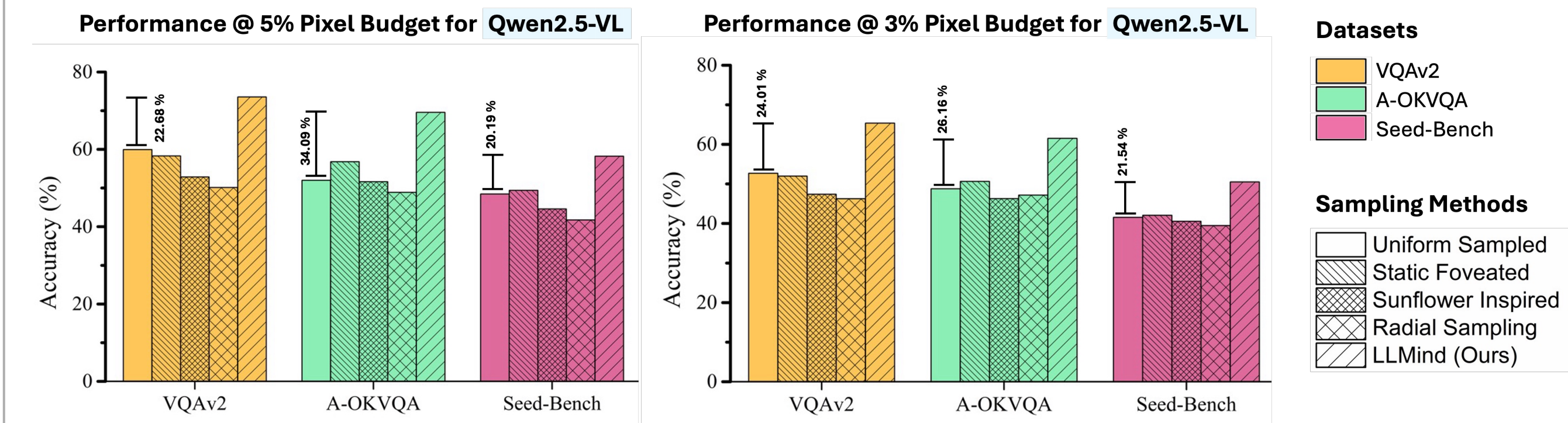
Non-Uniform Sampling



Experiments and Results

Q-1: Does LLMind improve scene-level VQA accuracy across diverse benchmarks ? [Yes]
Q-2: Can LLMind enhance region-guided VQA by reducing background distraction ? [Yes]
Q-3: What is the impact of the CSF on LLMind ? [CSF Improves Reasoning in VLMs]

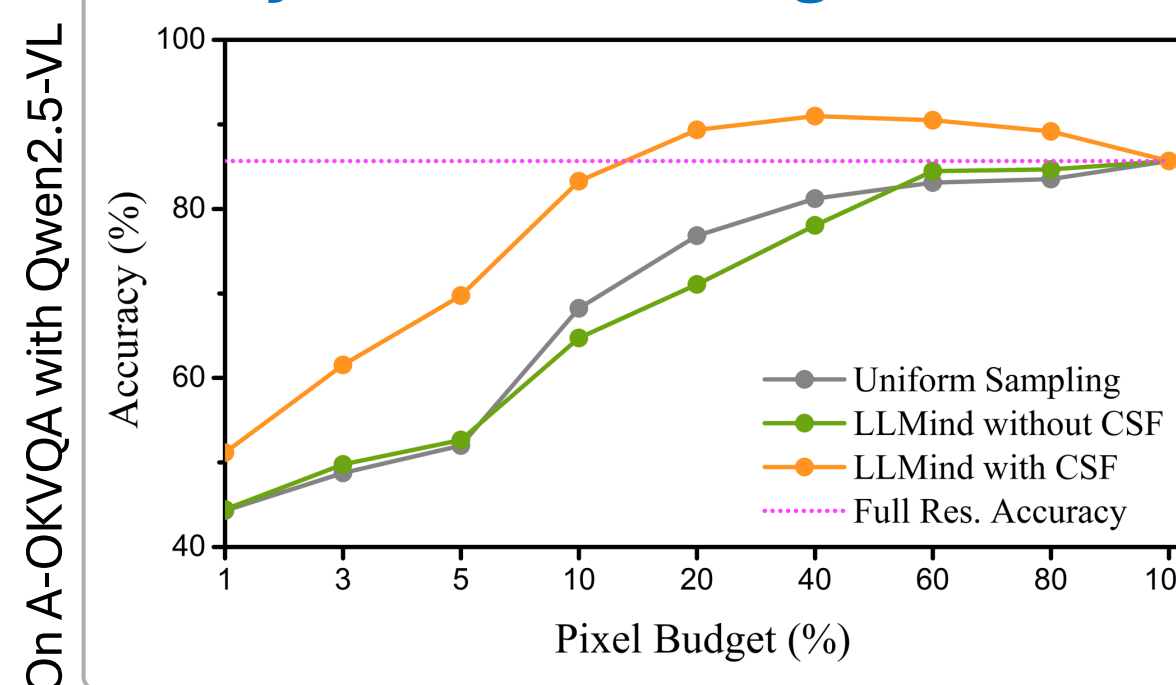
Quantitative Results for LLMind Across Various VQA Datasets



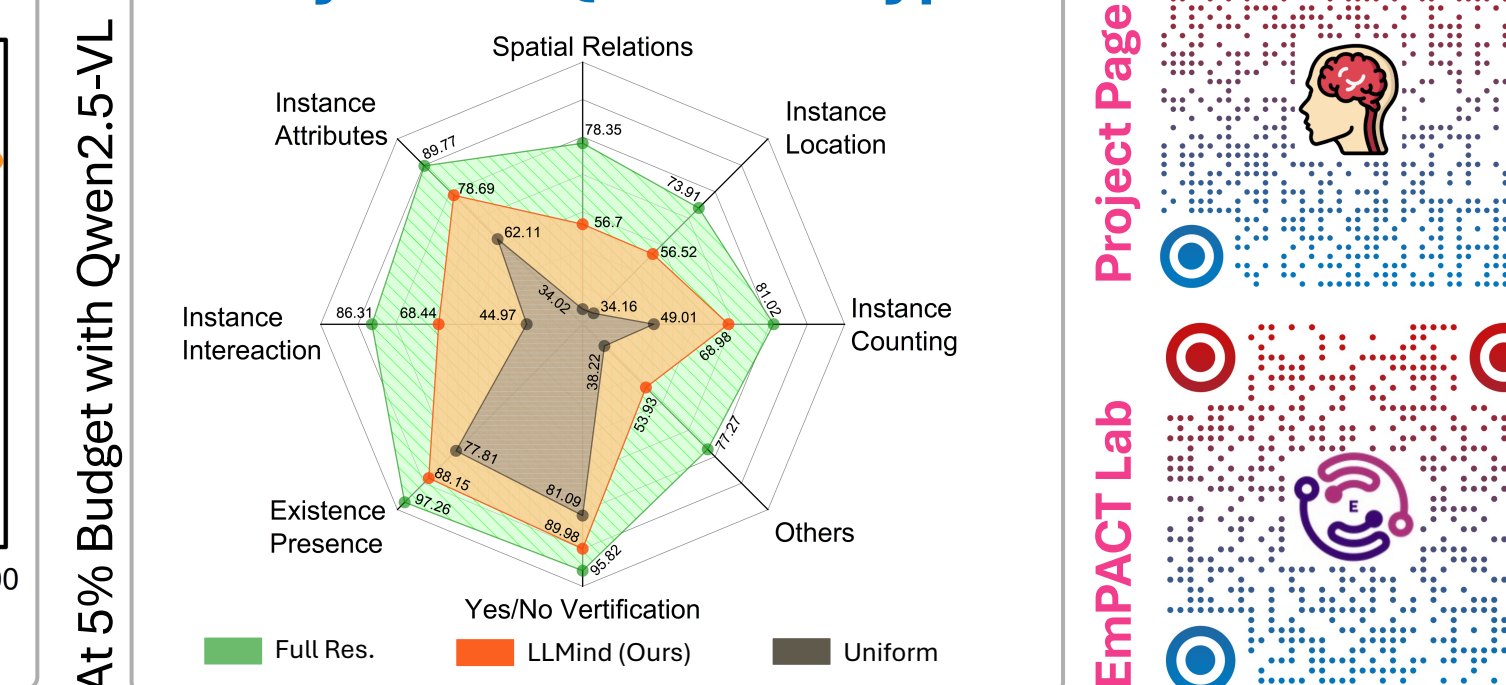
Qualitative Results for LLMind using Qwen2.5-VL at 5% Pixel Budget



Analysis on Pixel Budget and CSF



Analysis on Question Types



References

- [1] Gizdov, Andrey, Shimon Ullman, and Daniel Harari. "Seeing more with less: human-like representations in vision models." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025.
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- [5] Zhou, Sharon, et al. "Data augmentation with Mobius transformations." ML: Science and Technology (2021).

